Chapter 17 Notes

Section 1

Notes

Here is an explanation for the remarks given on page 224. Note that I am assuming M contains all ordinals (see the section for Remark 3). Definition 1.1 to even work in the first place. If a function f has domain α, and is such that, for all $\beta < \alpha$, $f(\beta) = R_\beta^{\mathcal{G}}$, call f an $R_\alpha^{\mathcal{G}}$ function. Let R-alpha func(f) stand for the following formula:

$$\text{Func}(f) \wedge \text{Dom}(f) = \alpha^+ \wedge f(0) = 0 \wedge$$

$$(\forall \beta \leq \alpha)(\beta^+ < \alpha \supset (\forall x)(x \in f(\beta^+) \equiv x \subseteq \mathcal{G} \times f(\beta)))$$

$$\wedge (\forall \beta < \alpha)(\text{beta is a limit ordinal} \supset (\forall x)(x \in f(\beta)$$

$$\equiv (\exists \delta < \beta)(x \in f(\delta))))$$

Note that R-alpha func(f) just means f is the $R_\alpha^{\mathcal{G}}$ function.

Lemma 0.1. For any α ($\alpha \in M$), if, $R_\beta^{\mathcal{G}} \in M$ for all $\beta < \alpha$, then both $R_\alpha^{\mathcal{G}} \in M$ and the $R_\alpha^{\mathcal{G}}$ function are in M.

Proof. By transfinite induction.

The least ordinal case is obvious (because $\emptyset \in M$).

Suppose it holds for all ordinals less than some ordinal α. So suppose that $R_\beta^{\mathcal{G}} \in M$ for all $\beta < \alpha$. Then it follows that the $R_\beta^{\mathcal{G}}$ function is in M for all $\beta < \alpha$. We first show that $R_\alpha^{\mathcal{G}} \in M$. Well, if α is a successor ordinal of some β, then $R_\alpha^{\mathcal{G}}$ is defined by the formula

$$(\forall y)(y \in x \equiv y \subseteq \mathcal{G} \times R_\beta^{\mathcal{G}})$$

which works, as $\mathcal{G} \times R_\beta^{\mathcal{G}} \in M$ for the following reason: both $\mathcal{G}$ and $R_\beta^{\mathcal{G}}$ are in M, meaning $\{\mathcal{G}, R_\beta^{\mathcal{G}}\} \in M$ (because M is a first order universe). Thus $\cup\{\mathcal{G}, R_\beta^{\mathcal{G}}\} \in M$, meaning $\mathcal{P}(\cup\{\mathcal{G}, R_\beta^{\mathcal{G}}\}) \cap M \in M$, and thus (1) $\mathcal{P}(\mathcal{P}(\cup\{\mathcal{G}, R_\beta^{\mathcal{G}}\}) \cap M) \cap M \in M$. Now, it is obvious that the formula in M in (1) is such that $\langle x, y\rangle$ is in it for each $x \in \mathcal{G}$ and each $y \in R_\beta^{\mathcal{G}}$. Then it is easy to see that $\mathcal{G} \times R_\beta^{\mathcal{G}}$ is first-order definable over M, and, because this would mean it is a first-order definable susbset of the set in (1), $\mathcal{G} \times R_\beta^{\mathcal{G}} \in M$ (because M is first-order swelled). Then $\mathcal{P}(\mathcal{G} \times R_\beta^{\mathcal{G}}) \cap M$, that is, $R_{\beta^+}^{\mathcal{G}}$, is in M.

Now we show that the $R_\alpha^{\mathcal{G}}$ function is definable over M. The $R_\alpha^{\mathcal{G}}$ function is definable over M by the formula

$$(x = \beta \wedge y = R_\beta^{\mathcal{G}}) \vee$$

$$(\exists \delta < \alpha)(\exists f)(\text{R-delta func}(f) \wedge x \in \text{Dom}(f) \wedge y = f(x))$$

Because $\alpha^+ \in M$, this means that the range of the $R_\alpha^{\mathcal{G}}$ function, the set S of all $R_\beta^{\mathcal{G}}$ for $\beta < \alpha$, is in M (and thus also $R_\alpha^{\mathcal{G}} \in M$). Hence $\{\alpha, S\} \in M$, $\cup\{\alpha, S\} \in M$, $\mathcal{P}(\cup\{\alpha, S\}) \in M$, and $\mathcal{P}(\mathcal{P}(\cup\{\alpha, S\})) \in M$. Now, for each $\beta < \alpha$ and $x \in S$, it of course follows that $\langle \beta, x\rangle \in \mathcal{P}(\mathcal{P}(\cup\{\alpha, S\}))$. That is, the $R_\alpha^{\mathcal{G}}$ function is a first-order definable subset of $\mathcal{P}(\mathcal{P}(\cup\{\alpha, S\}))$, which is in M, meaning the function itself is in M.

Now suppose α is a limit ordinal. Then the $R_\alpha^{\mathcal{G}}$ function is defined by

$$(\exists \delta < \alpha)(\exists f)(\text{R-delta func}(f) \wedge x \in \text{Dom}(f) \wedge y = f(x)) \vee$$

$$(x = \alpha \wedge y = R^{\mathcal{G}}_{\alpha})$$

and similar reasoning as in the above paragraph shows that the $R^{\mathcal{G}}_{\alpha}$ function is then in M. Then, because $R^{\mathcal{G}}_{\alpha}$ is just the union of the range of the $R^{\mathcal{G}}_{\alpha}$ function, it follows that $R^{\mathcal{G}}_{\alpha} \in M$.

□

Remark 1

The fact that each $R^{\mathcal{G}}_{\alpha} \in M$ is immediate from the above lemma (using an obvious transfinite induction argument). Then $\mathcal{D}^{\mathcal{G}}$ being a subclass of M follows because M is transitive.

Remark 2

We show that $R^{\mathcal{G}}_{\alpha} \subseteq R^{\mathcal{G}}_{\alpha+1}$. In fact, we directly prove the stronger result, that $\beta < \alpha$ implies $R^{\mathcal{G}}_{\beta} \subseteq R^{\mathcal{G}}_{\alpha}$.

Proof. By transfinite induction.

The least ordinal case is vacuous.

Suppose the result holds for some α. Now, if $x \in R^{\mathcal{G}}_{\alpha}$, there is some $\beta < \alpha$ such that all members of x are of the form $\langle y, z\rangle$ where $y \in \mathcal{G}$ and $z \in R^{\mathcal{G}}_{\beta}$. By the induction assumption, this means that $z \in R^{\mathcal{G}}_{\alpha}$. Then $\langle y, z\rangle$ would also be in $R^{\mathcal{G}}_{\alpha}$ for all such y and z, and then so would x (as we are assuming x is in M).

The limit ordinal case is obvious. □

Remark 3

We first prove the following lemma.

Lemma 0.2. For each ordinal α, if α is the least ordinal or a limit ordinal, $R^{\mathcal{G}}_{\alpha} \in R^{\mathcal{G}}_{\alpha^+}$ (note, however, that $R^{\mathcal{G}}_{\alpha^+} \notin R^{\mathcal{G}}_{\alpha^{++}}$ in general for successor ordinals α^+).

Proof. The least ordinal case is obvious, as $R^{\mathcal{G}}_0 \in R^{\mathcal{G}}_1$ because $R^{\mathcal{G}}_0 = \emptyset$ and $R^{\mathcal{G}}_1 = \{\emptyset\}$.

Consider some limit ordinal λ. Hence $R^{\mathcal{G}}_{\lambda} = \cup_{\alpha<\lambda} R^{\mathcal{G}}_{\alpha}$. We already know that $R^{\mathcal{G}}_{\lambda}$ is in M (by (1) of the remarks). Also, it can easily be seen that $R^{\mathcal{G}}_{\lambda} \subseteq \mathcal{G} \times R^{\mathcal{G}}_{\lambda}$. Thus, by definition of $R^{\mathcal{G}}_{\lambda^+}$, it follows that $R^{\mathcal{G}}_{\lambda} \in R^{\mathcal{G}}_{\lambda^+}$. □

We now prove that $R^{\mathcal{G}}_{\alpha+1} \neq R^{\mathcal{G}}_{\alpha}$.

Proof. By transfinite induction. In particular, we show that, for every α, $R^{\mathcal{G}}_{\beta} \neq R^{\mathcal{G}}_{\alpha}$ for every $\beta < \alpha$.

Suppose the result holds for all ordinals less than some ordinal δ.

Least ordinal case: The case when $\delta = \emptyset$ is vacuous.

Successor ordinal case: Suppose δ is the successor of some ordinal α. There are two cases: either α itself is a successor ordinal, or it is a limit ordinal (of course, α could be $\emptyset$, but that case is trivial). We must show the result holds for α^+ (that is, δ). Note that it suffices to show that $R^{\mathcal{G}}_{\alpha+1} \neq R^{\mathcal{G}}_{\alpha}$, for, if we show that, then if $R^{\mathcal{G}}_{\alpha+1} = R^{\mathcal{G}}_{\beta}$ for some $\beta < \alpha$, then it would follow that $R^{\mathcal{G}}_{\beta} = R^{\mathcal{G}}_{\alpha}$ by part (2) of the remarks, contrary to the induction assumption.

On the one hand, suppose that α itself is a successor ordinal (say of some β). Thus, by the assumption, we know that there must be some $y \in R^{\mathcal{G}}_{\alpha}$ such that $y \notin R^{\mathcal{G}}_{\beta}$. Consider any $x \in \mathcal{G}$. Then $\{\langle x, y\rangle\} \in R^{\mathcal{G}}_{\alpha+1}$ (because M is a first-order universe). If $R^{\mathcal{G}}_{\alpha+1} = R^{\mathcal{G}}_{\alpha}$, then it would follow that $z = \{\langle x, y\rangle\}$ for some $z \in R^{\mathcal{G}}_{\alpha}$. What this would mean is that $\langle x, y\rangle \in \mathcal{G} \times R^{\mathcal{G}}_{\beta}$. But that would mean $y \in R^{\mathcal{G}}_{\beta}$, a contradiction. Thus it must be that $R^{\mathcal{G}}_{\alpha+1} \neq R^{\mathcal{G}}_{\alpha}$.

On the other hand, suppose that α is a limit ordinal. By Lemma 0.2, it follows that $R^{\mathcal{G}}_{\alpha} \in R^{\mathcal{G}}_{\alpha^+}$. If $R^{\mathcal{G}}_{\alpha^+} = R^{\mathcal{G}}_{\alpha}$, then $R^{\mathcal{G}}_{\alpha} \in R^{\mathcal{G}}_{\alpha}$, which contradicts the assumption that M, and, by extension, $R^{\mathcal{G}}_{\alpha}$, is well founded.

Limit ordinal case: This is easy. □

Now we show that $\mathcal{D}^{\mathcal{G}}$ is a proper class in M (that is, $\mathcal{D}^{\mathcal{G}} \notin M$). Suppose it were in M. Then, because $R^{\mathcal{G}}_0, R^{\mathcal{G}}_1, ..., R^{\mathcal{G}}_{\alpha}, ...$ is an ordinal hierarchy, it follows by Theorem 2.1 of Chapter 7 that there is some ordinal α such that $\mathcal{D}^{\mathcal{G}} \subseteq R^{\mathcal{G}}_{\alpha}$. By (2), $\mathcal{D}^{\mathcal{G}} \subseteq R^{\mathcal{G}}_{\alpha+1}$. Also $R^{\mathcal{G}}_{\alpha+1} \subseteq \mathcal{D}^{\mathcal{G}}$, meaning $R^{\mathcal{G}}_{\alpha+1} = \mathcal{D}^{\mathcal{G}}$. But this contradicts the fact that $R^{\mathcal{G}}_{\alpha+1} \neq R^{\mathcal{G}}_{\alpha}$. Thus it cannot be that $\mathcal{D}^{\mathcal{G}} \in M$.

The above paragraph explains why I am assuming contains all ordinals. For if it didn't, then we wouldn't even know the sequence from Definition 1.1 was defined for all ordinals, and thus we couldn't use Theorem 2.1 of Chapter 7. However, it may indeed be that we could prove the result without this theorem - I was, however, unable to see this.

Remark 4

Suppose $f \subseteq \mathcal{G} \times \mathcal{D}^{\mathcal{G}}$ and f is an *M-set*. Now, for each $x \in f$, x is of the form $\langle y, z\rangle$ for some $y \in \mathcal{G}$ and some $z \in R^{\mathcal{G}}_{\alpha}$ for some α. Let $o(f)$ be the class of all α such that $\langle x, y\rangle \in f$ for at least one $y \in R^{\mathcal{G}}_{\alpha}$. We show that there must be some β higher than all the $\alpha \in o(f)$. For suppose there weren't. Then it would follow that $o(f)$ could be put into a 1-1 correspondence with On. But $o(f)$ is a set (because f is a set), which would mean that On is a set (by the axiom of substitution), which is impossible. Thus it must be that there is some β higher than all the $\alpha \in o(f)$. Consider the least such β. Because $f \in M$, it is easy to see that $f \in R^{\mathcal{G}}_{\beta+1}$, and thus $f \in \mathcal{D}^{\mathcal{G}}$.

Remark 5

I could not see how to show the isomorphism, but the rest of the remark seems to be just Exercise 1.2, which I will give here.

Here is how we could get this recursion to work in the first place. We define a function m_{α} for each α whose domain is $R^{\mathcal{G}}_{\alpha}$ and which satisfies the conditions of the function. We use transfinite recursion, where $m_{\alpha+1} = g(m_{\alpha})$, where $g(m_{\alpha})$ is the set containing all members of m_{α}, as well as all sets of the form $\langle x, \{m_{\alpha}(g) | \langle p, g\rangle \in x\}\rangle$ for $x \in R^{\mathcal{G}}_{\alpha+1} - R^{\mathcal{G}}_{\alpha}$.

Suppose $\mathcal{G} = \{p\}$, and a mapping m is defined by

$$m(f) = \{m(g) | \langle p, g\rangle \in f\}$$

First, we show that the mapping is 1-1. Note for this that an obvious transfinite induction argument shows that if $\beta < \alpha$, $m(g)$ stays constant between m applied to $R^{\mathcal{G}}_{\beta}$ and $R^{\mathcal{G}}_{\alpha}$ (that is, between m_{β} and m_{α}), for any $g \in R^{\mathcal{G}}_{\beta}$.

Proof. By transfinite induction.

The case for $R^{\mathcal{G}}_0$ obviously holds.

Suppose it holds for all ordinals less than some α. Now consider any f and h in $R^{\mathcal{G}}_{\alpha}$. Suppose $m(f) = m(h)$. Then

$$m(f) = \{m(g) | \langle p, g\rangle \in f\} = m(h) = \{m(l) | \langle p, l\rangle \in f\}$$

Then it easily follows by the induction assumption that $f = h$, as the values of $m(g)$ for all $g \in R^{\mathcal{G}}_{\beta}$ for some $\beta < \alpha$ stay constant. □

Next we show that, for all $f \in R_\alpha^{\mathcal{G}}$, $m(f) \in R_\alpha$.

Proof. By transfinite induction.

The case for $R_0^{\mathcal{G}}$ vacuously holds.

Suppose it holds for some α. Consider any $x \in R_{\alpha+1}^{\mathcal{G}}$; then $m(f) = \{m(g) | \langle p, g\rangle \in f\}$, and, by assumption, each $m(g)$ would be in R_α. Thus $m(f)$ is a subset of R_α, and, consequently, it is in $R_{\alpha+1}$ (because $R_{\alpha+1} = \mathcal{P}(R_\alpha)$.

The limit ordinal case is obvious. □

The last part of the exercise, showing that if $x \in R_\alpha$, then there is some $y \in R_\alpha^{\mathcal{G}}$ such that $m(y) = x$, was one the one I was unable to show.

Clarification

Definition 1.4 in the book needs clarification. In particular, the $[\![]\!]$ translation is being used, yet this has not yet been defined for the $\approx_\alpha$ relation. The definition should be: $[\![f \approx_{\alpha+1} g]\!] = f \approx_{\alpha+1} g$, $[\![f \approx_\lambda g]\!] = \Box\Diamond f \approx_\lambda$ for λ a limit ordinal, and $[\![x \epsilon y]\!] = \Box\Diamond x \epsilon y$.

The errata listed at `https://id144254.securedata.net/melvinfitting/errata/book_errors/RevisedSetBookErrors/RevisedSetCorrections.pdf` are important to note for this section.

It should be noted why $(f \approx_\lambda g) \supset [\![f \approx_\lambda g]\!]$ for limit ordinals λ. We prove this using transfinite induction. Suppose it holds for all ordinals less than λ. By the definition of $\approx_\lambda$, $(f \approx_\alpha g) \supset (f \approx_\lambda g)$ is valid in the model for all $\alpha < \lambda$. Thus $\Box\Diamond(f \approx_\alpha g) \supset \Box\Diamond(f \approx_\lambda g)$ is also valid in the model, that is, $[\![f \approx_\alpha g]\!] \supset [\![f \approx_\lambda]\!]$ is valid in the model. Now if $p \vDash (f \approx_\lambda g)$, then $p \vDash (f \approx_\alpha g)$ for some $\alpha < \lambda$, and by the induction assumption, this means $p \vDash [\![f \approx_\alpha g]\!]$, hence $p \vDash [\![f \approx_\lambda g]\!]$.

Proposition 1.5

One such error listed in the errata is with regards to the limit ordinal case in this proposition. The definition of $\approx_\lambda$ is not that $p \vDash [\![f \approx_\lambda]\!]$ implies $p \vDash [\![f \approx_\alpha]\!]$ for some $\alpha < \lambda$, but rather that $p \vDash f \approx_\lambda$ implies $p \vDash f \approx_\alpha$ for some $\alpha < \lambda$. Thus the inference made in the proof is invalid.

Instead, we show that $p \vDash f \approx_\lambda g$ implies $p \vDash [\![f \approx_\beta g]\!]$. Well, if $p \vDash f \approx_\lambda g$, then $p \vDash f \approx_\alpha g$ for some $\alpha < \lambda$. Then it easily follows that $p \vDash [\![f \approx_\alpha g]\!]$, and thus, by the induction assumption, $p \vDash [\![f \approx_\beta g]\!]$.

Thus $p \vDash (f \approx_\lambda g) \supset [\![f \approx_\beta g]\!]$ is valid in the model. Hence, using standard $S4$ manipulation, $p \vDash \Box\Diamond(f \approx_\lambda g) \supset \Box\Diamond[\![f \approx_\beta g]\!]$, that is, $p \vDash [\![f \approx_\lambda g]\!] \supset [\![f \approx_\beta g]\!]$, is valid in the model.

Exercise 1.3

We show that for $f, g \in R_\alpha^{\mathcal{G}}$, if $p \vDash (f \approx g)$, then $p \vDash (f \approx_\alpha g)$.

Proof. By transfinite induction. In particular, we make the induction assumption that, for all possible worlds p (we are not just considering one particular one), if $p \vDash (f \approx g)$ for $f, g \in R_\alpha^{\mathcal{G}}$, then $p \vDash (f \approx_\alpha g)$.

This vacuously holds for the case of 0, and the limit ordinal case is obvious.

The successor ordinal case is where the real work is required. Suppose the result holds for all ordinals less than some successor ordinal α^+. Suppose $f, g \in R_{\alpha^+}^{\mathcal{G}}$, and $p \vDash (f \approx g)$. Now, this means $p \vDash (f \approx_\beta g)$ for some β. This means

$p \vDash (f \approx_{\delta^+} g)$ for some successor ordinal $\delta^+ \leq \beta$. Thus we have

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\delta x)] \wedge$$

$$(\forall x)[(x \epsilon g) \supset (\exists y)(y \epsilon f \wedge y \approx_\delta x)]]\!]$$

meaning we have

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\delta x)]]\!]$$

which means we have, for each $d \in \mathcal{D}^{\mathcal{G}}$,

$$p \vDash [\![(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\delta x)]\!]$$

meaning we have

$$(1)\ p \vDash \Box\Diamond([\![x \epsilon f]\!] \supset [\![(\exists y)(y \epsilon g \wedge y \approx_\delta d))]\!]$$

Because we want to show $p \vDash (f \approx_{\alpha^+} g)$, what we aim to show is

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)] \wedge$$

$$(\forall x)[(x \epsilon g) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$$

It suffices to show both $p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)]]\!]$ and $p \vDash [\![(\forall x)[(x \epsilon g) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$. We give the proof of (2) $p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)]]\!]$ - the other case is similar. Now, (2) is the same as saying $p \vDash (\forall x)[\![(x \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha x)]\!]$. It suffices to show that, for all $d \in \mathcal{D}^{\mathcal{G}}$, $p \vDash [\![(d \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha d)]\!]$. So consider any $d \in \mathcal{D}^{\mathcal{G}}$. We must show that $p \vDash [\![(d \epsilon f) \supset (\exists y)(y \epsilon g \wedge y \approx_\alpha d)]\!]$, which is equivalent to $p \vDash \Box\Diamond([\![d \epsilon f]\!] \supset [\![(\exists y)(y \epsilon g \wedge y \approx_\alpha d)]\!])$, which is equivalent to $(2')\ p \vDash \Box\Diamond([\![d \epsilon f]\!] \supset \Box\Diamond(\exists y)[\![y \epsilon g \wedge y \approx_\alpha d]\!])$. To show this, consider any p' accessible from p. By (1), there is some p'' accessible from p' such that $p'' \vDash [\![x \epsilon f]\!] \supset [\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$, which is equivalent to (3) $p'' \vDash [\![d \epsilon f]\!] \supset \Box\Diamond[\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$. We show that (4) $p'' \vDash [\![d \epsilon f]\!] \supset \Box\Diamond(\exists y)[\![y \epsilon g \wedge y \approx_\alpha d]\!]$ (which would imply $(2')$). So suppose that $p'' \vDash [\![d \epsilon f]\!]$. By (3), this means that (5) $p'' \vDash \Box\Diamond[\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$. To show that (4) holds, consider any q accessible from p''. By (5), there is some q' accessible from q such that $q' \vDash [\![(\exists y)(y \epsilon g \wedge y \approx_\delta d)]\!]$. Then there is some q'' accessible from q' such that there is some $k \in \mathcal{D}^{\mathcal{G}}$ with $q'' \vDash [\![k \epsilon g \wedge k \approx_\delta d]\!]$, which implies both $q'' \vDash [\![k \epsilon g]\!]$ and $q'' \vDash [\![k \approx_\delta d]\!]$. Note that $d, k \in R^{\mathcal{G}}_\alpha$. Because $q'' \vDash [\![k \approx_\delta d]\!]$, it follows that $q'' \vDash \Box\Diamond(k \approx_\delta d)$. Thus there is some r' accessible from some r accessible from q'' such that $r' \vDash (k \approx_\delta d)$. Now, this implies $r' \vDash (k \approx d)$. By the induction assumption, this tells us that $r' \vDash (k \approx_\alpha d)$, which implies $r' \vDash [\![k \approx_\alpha]\!]$. Also, because r is accessibe from q'' (by the transitivity of the accessibility relation), and $q'' \vDash [\![k \epsilon g]\!]$, $r' \vDash [\![k \epsilon g]\!]$. Now, because r' is accessible from q, the fact that we have shown both $r' \vDash [\![k \approx_\alpha]\!]$ and $r' \vDash [\![k \epsilon g]\!]$ proves that $p'' \vDash \Box\Diamond(\exists y)[\![(y \epsilon g) \wedge (y \approx_\alpha d)$, which completes the proof in this case. $\square$

Defintion 1.7

It was not entirely clear to me how $[\![f \in g]\!]$ was meant to be defined, but it seems to me like it would just consist of adding $\Box\Diamond$ to the beginning of Definition 1.7.

Section 2

Lemma 2.1

It is claimed that $(f \approx_\alpha g) \equiv [\![f \approx_\alpha g]\!]$, yet this is incorrect. In particular, if $[\![(f \approx_\alpha g)]\!]$ for α a limit ordinal, it does not necessarily follow that $(f \approx_\alpha g)$.

The proof of this lemma leaves out the limit ordinal case. Here is an explanation for that case. Suppose the result holds for all $\alpha < \lambda$ for some limit ordinal λ. Now suppose that $p \vDash [\![f \approx_\lambda g]\!]$ and $p \vDash [\![g \approx_\lambda h]\!]$, which imply both (1) $p \vDash \Box\Diamond(f \approx_\lambda g)$ and (2) $p \vDash \Box\Diamond(g \approx_\lambda h)$. To prove $p \vDash [\![f \approx_\lambda h]\!]$, it suffices to show that $p \vDash \Box\Diamond(f \approx_\lambda h)$. So consider any p' accessible from p. By (1), this means there is some p'' accessible from p' such that $p'' \vDash f \approx_\lambda g$, meaning $p'' \vDash f \approx_\alpha g$ for some $\alpha < \lambda$, and thus (3) $p'' \vDash [\![f \approx_\alpha g]\!]$. Because p'' is itself accessible from p, it follows by (2) that there is some p''' accessible from p'' such that $p''' \vDash g \approx_\lambda h$, which implies $p''' \vDash g \approx_\beta h$ for some $\beta < \lambda$, which implies $p''' \vDash [\![g \approx_\beta h]\!]$. Now, by (3), and the fact that p''' is accessible from p'', $p''' \vDash [\![f \approx_\alpha g]\!]$. Pick the greater of α and β (say it is α). Then, by cumulativity, $p''' \vDash [\![g \approx_\alpha h]\!]$. Hence, by the induction assumption, it follows that $p''' \vDash [\![f \approx_\alpha h]\!]$. What this means is that for some q' accessible from some q accessible from p''', $q' \vDash f \approx_\alpha h$, which implies $q' \vDash f \approx_\lambda h$ (because $\alpha < \lambda$). But q' is accessible from p'. Thus we have shown, for every world p' accessible from p, there is some world q' such that $q' \vDash f \approx_\lambda h$ holds, and this proves $p \vDash \Box\Diamond(f \approx_\lambda h)$, that is, $p \vDash [\![f \approx_\lambda h]\!]$.

Theorem 2.2

The proof of this theorem is incorrect. It is claimed that $p \vDash [\![f \approx g]\!]$ implies $p \vDash [\![f \approx_\alpha g]\!]$ for some α, but this may not be the case. Here is a corrected proof (it is in fact reminiscent of the proof of the limit ordinal case of Lemma 2.1).

Suppose $p \vDash [\![f \approx g]\!]$, that is, (1) $p \vDash \Box\Diamond(f \approx g)$. Also suppose $p \vDash [\![g \approx h]\!]$, that is, (2) $p \vDash \Box\Diamond(g \approx h)$. To show $p \vDash [\![f \approx h]\!]$, we show $p \vDash \Box\Diamond(f \approx h)$. So consider any p' accessible from p. Then, by (1), there is some p'' accessible from p' such that $p'' \vDash f \approx g$, meaning $p'' \vDash f \approx_\alpha g$ for some α, hence (3) $p'' \vDash [\![f \approx_\alpha g]\!]$. Because p'' is itself accessible from p, (2) tells us that $p''' \vDash g \approx h$ for some p''' accessible from p'', which means $p''' \vDash g \approx_\beta h$ for some β, which implies $p''' \vDash [\![g \approx_\beta h]\!]$. Because p''' is accessible form p'', we have $p''' \vDash [\![f \approx_\alpha g]\!]$ by (3). Without loss of generality, let α be the maximum of α and β. Then, by cumulativity, $p''' \vDash [\![g \approx_\alpha h]\!]$. Hence, by Lemma 2.1, it follows that $p''' \vDash [\![f \approx_\alpha h]\!]$. This means $q' \vDash f \approx_h h$ for some q' accessible from some q accessible from p''', which implies $q' \vDash f \approx h$. But q' is then accessible from p'. Thus we have shown that, for every p' accessible from p, there is some q' accessible from p' such that $q' \vDash f \approx h$, which proves $p \vDash \Box\Diamond(f \approx h)$, that is, $p \vDash [\![f \approx h]\!]$.

Exercise 2.1

We first prove that if $f \in R^{\mathcal{G}}_\alpha$, then $p \vDash (f \approx_\alpha f)$.

Proof. By transfinite induction. The least ordinal case is vacuous, and the limit ordinal case is obvious.

We now consider the successor ordinal case. So suppose the result holds for some α, and suppose $f \in R^{\mathcal{G}}_{\alpha^+}$. We must show $p \vDash (f \approx_{\alpha^+} f)$, that is,

$$p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)] \wedge$$
$$(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$$

which is the same as just showing $p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha x)]]\!]$. So consider any $d \in \mathcal{D}^{\mathcal{G}}$. We must show that $p \vDash [\![(d \epsilon f) \supset (\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!]$, that is, $p \vDash \Box\Diamond([\![d \epsilon f]\!] \supset [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!])$. So consider any p' with $p \mathcal{R} p'$. Then consider any p'' with $p' \mathcal{R} p''$. We show that $p'' \vDash ([\![d \epsilon f]\!] \supset [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!])$. Suppose $p'' \vDash [\![d \epsilon f]\!]$. This means that, for some q' accessible from some q accessible from p'', $q' \vDash d \epsilon f$, which implies that $d \in R^{\mathcal{G}}_\alpha$. Now, to show that $p'' \vDash [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!]$, we show that $p'' \vDash [\![(d \epsilon f \wedge d \approx_\alpha d)]\!]$, that is, that both $p'' \vDash [\![d \epsilon f]\!]$ and $p'' \vDash [\![d \approx_\alpha d]\!]$. We already know that $p'' \vDash [\![d \epsilon f]\!]$. As for $p'' \vDash [\![d \approx_\alpha d]\!]$, this is obvious now that we know $d \in R^{\mathcal{G}}_\alpha$ (because, by the induction assumption, this means that $r \vDash d \approx_\alpha d$ for all possible worlds r). Thus we have $p'' \vDash [\![(d \epsilon f \wedge d \approx_\alpha d)]\!]$, hence $p'' \vDash [\![(\exists y)(y \epsilon f \wedge y \approx_\alpha d)]\!]$, which is what we wanted to show. □

Thus, we know that $f \in R^{\mathcal{G}}_\alpha$ implies $p \vDash (f \approx_\alpha f)$, hence $p \vDash (f \approx f)$, hence $p \vDash [\![(f \approx f)]\!]$.

Lemma 2.6

There is a problem in this proof. $[\![f \approx g]\!]$ being true at p does not mean $[\![f \approx_{\alpha+1} g]\!]$ is true at p for some α, but rather, $[\![f \approx_{\alpha+1} g]\!]$ is true at some q accessible from p. This problem is easily corrected.

Exercise 2.2

We prove that $[\![(\forall x)(\forall y)[(x \approx y) \supset (\phi(x) \equiv \phi(y))]]\!]$ is $S4$ valid in $\mathcal{M}$. It suffices to show that, for all $f, g \in \mathcal{D}^{\mathcal{G}}$, and any possible world p, $p \vDash [\![(d \approx f) \supset (\phi(d) \equiv \phi(f))]\!]$, that is, $p \vDash \Box\Diamond([\![d \approx f]\!] \supset [\![(\phi(d) \equiv \phi(f))]\!])$. So consider any p' with $p \mathcal{R} p'$ and any p'' with $p' \mathcal{R} p''$. We show $p'' \vDash ([\![d \approx f]\!] \supset [\![(\phi(d) \equiv \phi(f))]\!])$. Suppose $p'' \vDash [\![d \approx f]\!]$. For the sake of contradiction, suppose that $p'' \nvDash [\![\phi(d) \equiv \phi(f)]\!]$. This would mean either $p'' \nvDash [\![\phi(d) \supset \phi(f)]\!]$ or $p'' \nvDash [\![\phi(f) \supset \phi(d)]\!]$. Without loss of generality, suppose it is the former case. Then, for some p''' accessible from p'', both $p''' \vDash [\![\phi(d)]\!]$ and $p''' \vDash [\![\sim\phi(f)]\!]$ hold. Now, because p''' is accessible from p'', and $p'' \vDash [\![d \approx f]\!]$, $p''' \vDash [\![d \approx f]\!]$. Thus, by Theorem 2.7, and the assummption that $\phi(x)$ has only x free, $p''' \vDash [\![\phi(d)]\!] \equiv [\![\phi(f)]\!]$. Thus, because we have $p''' \vDash [\![\phi(d)]\!]$, it follows that $p''' \vDash [\![\phi(f)]\!]$. But this contradicts $p''' \vDash [\![\sim\phi(f)]\!]$.

Theorem 2.10

Here is why Theorem 2.10 follows from Lemma 2.9. Suppose $p \vDash (\forall x)[\![x \in f \equiv x \in g]\!]$ and $f, g \in R_\alpha^{\mathcal{G}}$. Then, by Lemma 2.9, $p \vDash [\![f \approx_\alpha g]\!]$, which implies $p \vDash [\![f \approx_{\alpha^+} g]\!]$, which implies $p \vDash [\![f \approx_{\alpha^+} g]\!]$, which implies $p \vDash f \approx g$, which implies $p \vDash [\![f \approx g]\!]$.

Definition 3.1

It is not explained in the book, but here is why $\hat{x} \in \mathcal{D}^{\mathcal{G}}$ in the first place.

First, define the following ordinal hierarchy:

$R_0^M = \emptyset$

$R_{\alpha^+}^M = \mathcal{P}(R_\alpha^M) \cap M$

$R_\lambda^M = \cup_{\alpha<\lambda} R_\alpha^M$ for limit ordinals λ

Each R_α^M is just the set of all members of R_α that happen to be in M. An obvious transfinite induction argument (using the fact that M is a first-order universe) shows that each R_α^M is in M (for $\alpha \in M$). Let $R^M = \cup_\alpha R_\alpha$. By Theorem 3.3 of Chapter 8 (and the fact that M is transitive and well founded), each element of M has a rank. It follows that, in fact, $R^M = M$. Thus if we can show that there is a function satisfying the requirenments of Definition 3.1 for R^M, it also satisfies the requirenments for M itself. In particular, we want an f such that $f(x) = \{\langle p, f(y)\rangle | p \in \mathcal{G} \wedge y \in x\}$. To show this, we show that there is a function f_α for each α, whose domain is R_α^M, and such that $f_\alpha(x) = \{\langle p, f_\alpha(y)\rangle | p \in \mathcal{G} \wedge y \in x\}$. This is of course immediate by any of the transfinite recursion theorems.

Call a function an M_α function if it satisfies the following conditons: its domain is α, and for each α, $f(\alpha) = f_\alpha$.

Let M-func$_\alpha(x)$ stand for the following:

$$\text{Func}(x) \wedge \text{Dom}(x) = R_\alpha^M \wedge f(0) = 0 \wedge$$

$$(\forall \beta < \alpha)[\beta^+ < \alpha \supset$$

$$(\forall y)(y \in f(\beta^+) \equiv (y \in f(\beta) \vee (\exists v)(\exists w)(y = \langle w, v\rangle \wedge$$

$$w \in (R_{\beta^+}^M - R_\beta^M) \wedge (\forall z)(z \in v \equiv (\exists p)(\exists s)(p \in \mathcal{G} \wedge s \in w \wedge z = \langle p, f(\beta)(s)\rangle)))))]$$

$$\wedge(\forall \lambda < \alpha)[\text{lambda is a limit ordinal} \supset (\forall y)(y \in f(\lambda) \equiv (\exists \beta < \lambda)(y \in f(\beta)))])$$

If the M_α function happens to be in M, then M-func$_\alpha(f)$ will be true in M for that function f.

We first prove that each f_α is in M (for all α in M), and each M_α function is in M.

Theorem 0.3. For all $\alpha \in M$, $f_\alpha \in M$, and the M_α function is in M.

Proof. By transfinite induction.

The least ordinal case is given by the empty set axiom holding over M.

Suppose the result holds for some α. Thus we know that $f_\alpha \in M$ and the M_α function is in M. Let M_α stand for the M_α function. The f_{α^+} function is defined over M by:

$$(x \in \text{Dom}(f_\alpha) \wedge f_\alpha(x) = y) \vee$$

$$[x \in (R_{\alpha^+}^M - R_\alpha^M) \wedge (\forall z)(z \in y \equiv (\exists w)(\exists v)(z = \langle w, v\rangle \wedge$$

$$w \in \mathcal{G} \wedge (\exists s)(s \in x \wedge v = f_\alpha(s))))]$$

(note that this relies upon each such y being in M, but this is easy to see using the induction assumption, because it is a first-order definable subset of f_α) Because the domain of the f_{α^+} function, $R_{\alpha^+}^M$, is in M, it follows by the axiom of substitution

(over M) that the range is also in M. Then the set $S : \mathcal{P}(\mathcal{P}(\cup\{\mathrm{Dom}(f_{\alpha^+}), \mathrm{Ran}(f_{\alpha^+})\}) \cap M) \cap M$ is in M. Then, because f_{α^+} is a first-order definable (over M) subset of S, it follows that $f_{\alpha^+} \in M$.

Next we show that $M_{\alpha^+} \in M$. This function is definable by

$$(x \in \mathrm{Dom}(M_\alpha) \wedge M_\alpha(x) = y) \vee (x = \alpha \wedge y = f_\alpha)$$

and because the domain of the function, α^+, is in M, it follows that so is the range. Then by an argument similar in nature to the one in the previous paragraph, it follows that $M_{\alpha^+} \in M$

Now we consider the limit ordinal case. Suppose the result holds for all $\alpha < \lambda$ for some limit ordinal λ. First, the M_λ function is definable over M by

$$(\exists \alpha < \lambda)(\exists f)(\text{M-func}_\alpha(f) \wedge x \in \mathrm{Dom}(f) \wedge f(x) = y)$$

Because the domain of this function, λ, is in M, it follows that so is the range, that is, the set S of all f_α for $\alpha < \lambda$. But then $\cup S$, that is, f_λ, is in M. Also, because both λ and S are in M, it easily follows that the M_λ function is also in M.

□

Because we have shown each f_α to be in M, it follows by the transitivity of M that $f_\alpha(x) \in M$ for each $x \in R^M_\alpha$. Next, we show that if $x \in R^M_\alpha$, then $\hat{x} \in R^{\mathcal{G}}_\alpha$, that is, $f_\alpha(x) \in R^{\mathcal{G}}_\alpha$.

Proof. By transfinite induction.

The least ordinal case is vacuous, and the limit ordinal case is obvious.

Suppose the result holds for some α, that is, for each $x \in R^M_\alpha$, $f_\alpha \in R^{\mathcal{G}}_\alpha$. Now consider any $x \in R^M_{\alpha^+}$. We already have shown that $f_{\alpha^+}(x) \in M$. Thus, if we can show that all members of $f_{\alpha^+}(x)$ are of the form $\langle p, y\rangle$ for some $y \in R^{\mathcal{G}}_\alpha$ and some $p \in \mathcal{G}$, the desired result follows. Well, by definition of the f_{α^+} function, each member of $f_{\alpha^+}(x)$ is of the form $\langle p, f_\alpha(y)\rangle$ for some $y \in x$. But because $y \in x$, and $x \in R^M_{\alpha^+}$, it follows that $y \in R^M_\alpha$, and thus, by the induction assumption, $f_\alpha(y) \in R^{\mathcal{G}}_\alpha$, which completes the proof.

□

Lastly, we show that x has rank α (for $x \in M$) iff $\hat{x}$ has $\mathcal{G}$-rank α.

Proof. By transfinite induction. The least ordinal case is obvious.

Suppose the result holds for all ordinals less than α (where $\alpha > 0$).

On the one hand, suppose that x has rank α. Thus $x \in R^M_{\alpha^+} - R^M_\alpha$. We have already shown that this means $\hat{x} \in R^{\mathcal{G}}_{\alpha^+}$. Suppose that $\hat{x} \in R^M_\alpha$. Then, for some successor ordinal $\beta^+ \leq \alpha$, it follows that $\hat{x} \in R^M_{\beta^+}$, and thus all elements of $\hat{x}$ are of the form $\langle p, \hat{y}\rangle$ where $\hat{y} \in R^{\mathcal{G}}_\beta$ and $y \in x$. This would mean that each such $\hat{y}$ has $\mathcal{G}$-rank less than β, which, by the induction assumption, means each $y \in x$ is in R^M_β, which would mean $x \in R^M_{\beta^+}$, which is a contradiction, as $\beta < \alpha$, but x was assumed to have rank α.

On the other hand, suppose that $\hat{x}$ has $\mathcal{G}$-rank α. By the definition of $R^{\mathcal{G}}_{\alpha^+}$ and $\hat{x}$, this implies that $x \in R^M_{\alpha^+}$ (by the induction assumption). Suppose that $x \in R^M_\alpha$. Then, for some $\beta^+ \leq \alpha$, it follows that $x \in R^M_{\beta^+}$ and each element of x is in R^M_β (has rank less than β). Hence, by the induction assumption, it follows that, for each $y \in x$, $\hat{y} \in R^{\mathcal{G}}_\beta$, which would mean (by definition of $R^{\mathcal{G}}_{\beta^+}$) that $\hat{x} \in R^{\mathcal{G}}_{\beta^+}$, which is a contradiction, as $\beta < \alpha$, but $\hat{x}$ was assumed to have $\mathcal{G}$-rank α.

□

Proposition 3.3

It is said in the proof of this proposition that we can assume ϕ has all negations at the atomic level, as all Σ_0 formulas are equivalent to one in this form. Note that this follows from on obvious induction argument on the degree of ϕ.

"But then $p \vDash (a \epsilon \hat{t})$" is not correct. Rather, it follows that $p' \vDash (a \epsilon \hat{t})$ for some p' accessible from p (though this is not a major issue in the proof).

Corollary 3.4

Note that we can use Proposition 3.3 because $t_1, ..., t_k$ must all be well founded, as it is assumed M is well founded and transitive.

Exercise 3.1

We prove that, if $p \vDash (\hat{a} \approx \hat{b})$, then a and b are the same set (note that the converse of this statement holds by the reflexivity of the $\approx$ relation). We prove this by induction on the α for which $p \vDash (\hat{a} \approx_\alpha \hat{b})$ holds.

The least ordinal case is vacuous, and the limit ordinal case is obvious.

Now consider the successor ordinal case. So suppose $p \vDash (\hat{a} \approx_{\alpha^+} \hat{b})$, and the result holds for α. We then have both (1) $p \vDash [\![(\forall x)(x \epsilon \hat{a} \supset (\exists y)(y \epsilon \hat{b} \wedge x \approx_\alpha y))]\!]$ and (2) $p \vDash [\![(\forall x)(x \epsilon \hat{b} \supset (\exists y)(y \epsilon \hat{a} \wedge x \approx_\alpha y))]\!]$. We show that a and b are the same set by showing that $a \subseteq b$ and $b \subseteq a$.

On the one hand, suppose that $c \in a$. Then it follows that, for every possible world q, $\langle q, \hat{c} \rangle \in \hat{a}$. Thus we have $p \vDash [\![\hat{c} \epsilon \hat{a}]\!]$, meaning, by (1), there is some p' with $p \mathcal{R} p'$ and some $d \in \mathcal{D}^{\mathcal{G}}$ such that $p' \vDash [\![(d \epsilon \hat{b} \wedge d \approx_\alpha \hat{c})]\!]$. Because $p' \vDash [\![(d \epsilon \hat{b})]\!]$, d is $\hat{e}$ for some $e \in b$. Thus $p' \vDash [\![\hat{e} \approx_\alpha \hat{c}]\!]$, meaning, for some p'' with $p' \mathcal{R} p''$, $p'' \vDash \hat{e} \approx_\alpha \hat{c}$, and this implies $e = c$ by the induction assumption. Thus, because $e \in b$, $c \in b$, and therefore $a \subseteq b$.

The other hand is similar, but using (2) instead of (1).

Exercise 3.2

(1) We first prove the universal bounded quantifier case. In this case, the formula is of the form $(\forall x \in s)\phi(x, s, s_1, ..., s_n)$.

On the one hand, suppose $(\forall x \in s)\phi(x, s, s_1, ..., s_n)$ is true in M, that is, (1) $(\forall x)(x \in s \supset \phi(x, s, s_1, ..., s_n))$. Now, suppose there was some p such that $p \nvDash [\![(\forall x \in \hat{s})\phi(x, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. This would mean, for some $d \in \mathcal{D}^{\mathcal{G}}$, $p \nvDash [\![(d \in \hat{s}) \supset \phi(d, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. This means, for some p' with $p \mathcal{R} p'$, $p' \vDash [\![d \in \hat{s}]\!]$ and $p' \vDash [\![\sim \phi(d, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. Because $p' \vDash [\![d \in \hat{s}]\!]$, there must be some b and some p'' accessible from p' such that $p'' \vDash \hat{b} \in \hat{s}$ and $p'' \vDash [\![d \approx \hat{b}]\!]$. By the induction assumption, this means $b \in s$, and thus, by (1), $\phi(b, s, s_1, ..., s_n)$, and then, again by the induction assumption, $p'' \vDash [\![\phi(\hat{b}, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. Because $p'' \vDash [\![d \approx \hat{b}]\!]$, it follows by Corollary 2.8 that $p'' \vDash [\![\phi(d, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. But because p'' is accessible from p', and $p' \vDash [\![\sim \phi(d, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$, we have $p'' \vDash [\![\sim \phi(d, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$, a contradiction. Thus it must be that $[\![(\forall x \in \hat{s})\phi(x, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$ is true at every world of $\mathcal{G}$.

On the other hand, suppose that $p \vDash [\![(\forall x \in \hat{s})\phi(x, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. For the sake of contradiction, suppose that there was some $d \in M$ such $(d \in s) \wedge \sim \phi(d, s, s_1, ..., s_n)$. Then, by the induction assumption, $[\![\sim((\hat{d} \in \hat{s}) \supset \phi(\hat{d}, \hat{s}, \hat{s_1}, ..., \hat{s_n}))]\!]$ is true at all possible worlds, which means $p \nvDash [\![(\hat{d} \in \hat{s}) \supset \phi(\hat{d}, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$, a contradiction.

(2) Next, we prove the existential bounded quantifier case. In this case, the formula is of the form $(\exists x \in s)\phi(x, s, s_1, ..., s_n)$.

On the one hand, suppose $(\exists x \in s)\phi(x, s, s_1, ..., s_n)$ is tre in M. Thus, for some $d \in M$, both $d \in s$ and $\phi(d, s, s_1, ..., s_n)$

are true in M, meaning $(d \in s) \wedge \phi(d, s, s_1, ..., s_n)$ is true in M. By the induction assumption, this means that, for every possible world p, $p \vDash [\![(\hat{d} \in \hat{s}) \wedge \phi(\hat{d}, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$, which implies $p \vDash [\![(\exists x \in \hat{s})\phi(x, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$.

On the other hand, suppose that $p \vDash [\![(\exists x \in \hat{s})\phi(x, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. Thus, for some p' with $p\mathcal{R}p'$, there is some $d \in \mathcal{D}^{\mathcal{G}}$ with $p' \vDash [\![(d \in \hat{s}) \wedge \phi(d, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$, and thus both (1) $p' \vDash [\![(d \in \hat{s})]\!]$ and (2) $p' \vDash [\![\phi(d, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. By (1), there is some p'' accessible from p' such that there is some b with $p'' \vDash [\![d \approx \hat{b}]\!]$ and $p'' \vDash \hat{b} \in \hat{s}$. By the induction assumption, this means $b \in s$ is true in M. Also, by (2) and Corollary 2.8, we then have $p'' \vDash [\![\phi(\hat{b}, \hat{s}, \hat{s_1}, ..., \hat{s_n})]\!]$. Hence, by the induction assumption, $\phi(b, s, s_1, ..., s_n)$ is true in M, meaning $(b \in s) \wedge \phi(b, s, s_1, ..., s_n)$ is true in M, hence $(\exists x \in s)\phi(x, s, s_1, ..., s_n)$ is true in M.

Lemma 4.6

It is claimed that A is a proper class over M, though this does not necessarily follow. For suppose there was only one possible world p, and say $a = \{\langle p, 0\rangle\}$. Now, this means $p \vDash [\![\emptyset \in a]\!]$, but also that $p \nvDash [\![c \in a]\!]$ for any $c \neq \emptyset$. Thus $A = \{\emptyset\}$, and, because $\{\emptyset\} \in M$, it follows that $A \in M$.

Exercise 4.1

Say $f, g \in R_\alpha^{\mathcal{G}}$. Define a set h as follows:

$$\langle p, a\rangle \in h \equiv p \in \mathcal{G} \text{ and } a \in R_\alpha^{\mathcal{G}} \text{ and } p \vDash [\![a \in \cup f]\!]$$

It is not obvious that $h \in R_{\alpha+1}^{\mathcal{G}}$, but this is true (see the "Sets in Theorems of Section 4" section). It is obvious that, for $a \in R_\alpha^{\mathcal{G}}$,

$$p \vDash (a \epsilon h) \equiv p \vDash [\![a \in \cup\{f, g\}]\!]$$

Note that, for all worlds p, both $[\![f \in \{f, g\}]\!]$ and $[\![g \in \{f, g\}]\!]$ are true at p (by the reflexivitiy of the $\approx$ relation). Then, because both $f, g \in R_\alpha^{\mathcal{G}}$, it follows by the definition of h that $\langle p, f\rangle$ and $\langle p, h\rangle$ are both in g for all possible worlds p, and an easy consequence of this is that $(f \in h)$ and $(g \in h)$ are true at all possible worlds.

Now suppose that for some p, $p \nvDash [\![(\forall x)(x \in h \equiv x \in \{f, g\})]\!]$; we derive a contradiction.

(1) Suppose for some a, $p \nvDash [\![a \in h \supset a \in \{f, g\}]\!]$ (this case is nearly identical to that of (1) in Theorem 4.1). Then there is some p' with $p\mathcal{R}p'$ such that $p' \vDash [\![a \in h]\!]$ and $p' \vDash [\![\sim(a \in \{f, g\})]\!]$. By definition, for some a' and some p'' with $p'\mathcal{R}p''$, $p'' \vDash [\![a \approx a']\!]$ and $p'' \vDash [\![a' \epsilon h]\!]$. But then, for some p''' with $p''\mathcal{R}p'''$, $p''' \vDash a' \epsilon h$, and it then follows by the definition of h that $p''' \vDash [\![a' \in \{f, g\}]\!]$. Then by substitutivity, $p''' \vDash [\![a \in \{f, g\}]\!]$, and this is impossible since $[\![\sim(a \in \{f, g\})]\!]$ is true at p', hence at p'''.

(2) Suppose for some a, $p \nvDash [\![a \in \{f, g\} \supset a \in h]\!]$. Then for some p' with $p\mathcal{R}p'$, $p' \vDash [\![a \in \{f, g\}]\!]$ and $p' \vDash [\![\sim(a \in h)]\!]$. Since $p' \vDash [\![a \in \{f, g\}]\!]$, $p' \vDash [\![a \approx f \vee a \approx g]\!]$, meaning, at some world p'' accessible from p', either $p'' \vDash [\![a \approx f]\!]$ or $p'' \vDash [\![a \approx g]\!]$ holds. Without loss of generality, suppose it is the former that is true, that is, $p'' \vDash [\![a \approx f]\!]$. By substitutivity, we would then know that $p'' \vDash [\![\sim(f \in h)]\!]$, which contradicts the fact that $(f \in h)$ is true at all possible worlds.

Exercise 4.2

Suppose $p \nvDash [\![(\forall y)(\forall z)(\exists w)(\forall x)(x \in w \equiv x \in \{y, z\})]\!]$. Then for some $f, g \in \mathcal{D}^{\mathcal{G}}$, (1) $p \nvDash [\![(\exists w)(\forall x)(x \in w \equiv x \in \{f, g\})]\!]$. By Theorem 4.3, there is some $h \in \mathcal{D}^{\mathcal{G}}$ such that $p \vDash [\![(\forall x)(x \in h \equiv x \in \{f, g\})]\!]$. But by (1), $p \nvDash [\![(\forall x)(x \in h \equiv x \in \{f, g\})]\!]$, a contradiction.

Exercise 4.3

Suppose $p \not\models [\![(\exists y)(y \in a) \supset (\exists y)(y \in a \wedge (\forall z)(z \in a \supset \sim(z \in y))]\!]$. Then, for some p' with $p\mathcal{R}p'$, $p' \models [\![(\exists y)(y \in a)]\!]$ and $p' \models [\![\sim(\exists y)(y \in a \wedge (\forall z)(z \in a \supset \sim(z \in y)))]\!]$. Thus, for some p'' with $p'\mathcal{R}p''$ and for some $g \in \mathcal{D}^{\mathcal{G}}$, we have $p'' \models [\![g \in a]\!]$ and (1) $p'' \models [\![\sim(\exists y)(y \in a \wedge (\forall z)(z \in a \supset \sim(z \in y)))]\!]$. By Lemma 4.6, there is some p''' with $p''\mathcal{R}p'''$ and some c such that (2) $p''' \models [\![c \in a]\!]$ and $p''' \models (\forall z)[\![(z \in a \supset \sim(z \in c))]\!]$, which is equivalent to $p''' \models [\![(\forall z)(z \in a \supset \sim(z \in c))]\!]$. By (1), $p''' \models [\![(c \in a) \supset \sim(\forall z)(z \in a \supset \sim(z \in c))]\!]$, meaning, by (2), $p''' \vdash [\![\sim(\forall z)(z \in a \supset \sim(z \in c))]\!]$, a contradiction.

Sets in Theorems of Section 4

In Theorem 4.1, it is claimed that it is obvious that $g \in R^{\mathcal{G}}_{\alpha+1}$, and this is similarly claimed for similar sets in many other proofs in section 4. In reality, the result does follow, but much work is required. In fact, the work required is just the work of section 5 of this chapter. Using the formulas from Theorem 5.4, it *is* obvious that the various sets in the various theorems of Section 4 are definable over M, and because each of them is a subset of some $R^{\mathcal{G}}_{\alpha}$, this would mean the set itself is in M (because M is first-order swelled).

Lemma 5.2

There is an issue with the proof of this lemma. First, it is claimed that $(x \approx_\alpha y)$ and $[\![(x \approx_\alpha y)]\!]$ are equivalent, but this is not the case. Instead of trying to express

$$p \models (\forall x)\Box\Diamond[\Box\Diamond(x \epsilon f) \supset \Box\Diamond(\exists y)(\Box\Diamond(y \epsilon g) \wedge (x \approx_\alpha y))]$$

we should be trying to express

$$p \models (\forall x)\Box\Diamond[\Box\Diamond(x \epsilon f) \supset \Box\Diamond(\exists y)(\Box\Diamond(y \epsilon g) \wedge \Box\Diamond(x \approx_\alpha y))]$$

To do this, change $F(\mathcal{A}, p, f, g)$ to be

$$(\forall x \in \mathcal{D})(\forall q \overleftarrow{\mathcal{R}}\, p)(\exists q' \overleftarrow{\mathcal{R}}\, p)\{(\forall r \overleftarrow{\mathcal{R}}\, q')(\exists r' \overleftarrow{\mathcal{R}}\, r)\langle r', x\rangle \in f \supset$$

$$(\forall r \overleftarrow{\mathcal{R}}\, q')(\exists r' \overleftarrow{\mathcal{R}}\, r)(\exists y \in \mathcal{D})[(\forall s \overleftarrow{\mathcal{R}}\, r')(\exists s' \overleftarrow{\mathcal{R}}\, s)(\langle s', y\rangle \in g) \wedge$$

$$(\forall s \overleftarrow{\mathcal{R}}\, r')(\exists s' \overleftarrow{\mathcal{R}}\, s)(\langle s', x, y\rangle \in \mathcal{A})\} \wedge$$

$$p \in \mathcal{G} \wedge \mathcal{D}(f) \wedge \mathcal{D}(g)$$

(the last line I added in to ensure that p, f, g are correctly set up)

Proposition 5.3

There are multiple issues with the proof of this proposition. The given formula for Equal(p, f, g) does not work as is, simply because we do not yet know if the function asserted to exist in M in that formula really is in M. I will go about showing this by a slightly different method, though the end result is the same.

First, define the following sequence by transfinite recursion:

$s_0 = \emptyset$

$s_{\alpha^+} = \{\langle y, z, w\rangle | \text{Next Equal}(s_\alpha, y, z, w) \text{ and } z, w \in R^{\mathcal{G}}_{\alpha^+}\}$

$s_\lambda = \cup_{\alpha<\lambda} s_\alpha$ for λ a limit ordinal.

The condition that makes this different from the one in the book is the requirenment that z, w be in $R^{\mathcal{G}}_{\alpha^+}$. The reason for this is as follows: $f \approx_{\alpha^+} g$ may be true at p when f or g has rank $\geq \alpha^+$. What this could mean is that the set S of *all* $\langle p, f, g\rangle$ such that $p \Vdash f \approx_{\alpha^+} g$ may not be a subset of $(\mathcal{G} \times R^{\mathcal{G}}_{\alpha^+} \times R^{\mathcal{G}}_{\alpha^+}) \cap M$. This is important - this means that S may not be a set; but for the formula to come out correctly, we need to know that S is set. Thus we simply restrict the domain of each s_{α^+} to only those ordered pairs $\langle p, f, g\rangle$ where $f, g \in R^{\mathcal{G}}_{\alpha^+}$. Then, once we shown each s_α to be in M, we will be ready to make a formula for the $\approx$ relation itself.

Call a function f an s_α function if it satisfies the following conditions: it has a domain of α, and for each $\beta < \alpha$, $f(\beta) = s_\beta$. What we now aim to show is that, for each α (where $\alpha \in M$), both s_α and the s_α function are in M.

Let s-alpha func(f) stand for the following (it is fairly similar to the Equal(p, g, f) formula given in the book):

$$\text{Func}(f) \wedge \text{Dom}(f) = \alpha \wedge$$

$$(\forall \beta < \alpha)[(\beta = 0 \supset f(\beta) = 0) \wedge$$

$$(\beta^+ < \alpha \supset (\forall y, z, w)(\langle y, z, w\rangle \in f(\beta^+) \equiv$$

$$(y \in \mathcal{G} \wedge z, w \in R^{\mathcal{G}}_{\beta^+} \wedge \text{Next Equal}(f(\beta), y, z, w)))) \wedge$$

$$(\text{Limit Ordinal}(\beta) \supset f(\beta) = \cup_{\gamma<\beta} f(\gamma))]$$

If the s_α function happens to be in M, then s-alpha func(f) would be true in M for that function f.

Theorem 0.4. For each α in M, both s_α and the s_α function are in M.

Proof. The least ordinal case holds by the empty set axiom.

In the successor ordinal case, suppose both s_α and the s_α function are in M. Then s_{α^+} is definable over M by

$$(\exists w)(\exists v)(x = \langle w, v\rangle \wedge w, v \in R^{\mathcal{G}}_{\alpha^+} \wedge$$

$$\text{Next Equal}(s_\alpha, p, x, y))$$

which defines s_{α^+} over M. Because $\mathcal{G}$ and $R^{\mathcal{G}}_{\alpha^+}$ are both in M, thus so is $S : \mathcal{G} \times R^{\mathcal{G}}_{\alpha^+} \times R^{\mathcal{G}}_{\alpha^+}$. Then, because s_{α^+} is a subset of S, it follows that s_{α^+} is a first-order definable subset of S, and thus, because M is first-order swelled, $s_{\alpha^+} \in M$. The s_{α^+} function being in M then easily follows.

Now consider the limit ordinal case (for some limit ordinal λ). Then the s_λ function is definable over M by the following:

$$(\exists f)(\exists \alpha < \lambda)(\text{s-alpha func}(f) \wedge x \in \text{Dom}(f) \wedge f(x) = y)$$

The domain of the function is λ, and thus, because $\lambda \in M$, it follows by familiar reasoning that both s_λ and the s_λ function are in M.

□

Thus we have the s_α function in M for each $\alpha \in M$. Now we need to account for the possibility that $f \approx_\alpha g$ but one of f and g has a $\mathcal{G}$-rank $\geq \alpha$. We actually won't explicitly define a formula just for the $\approx_\alpha$ relation, but rather, the $\approx$ relation in its entirety. Now, there must be *some* β for which both $f, g \in R^{\mathcal{G}}_\beta$. This suggests the following definition for Equal(p, f, g):

$$p \in \mathcal{G} \wedge \mathcal{D}(f) \wedge \mathcal{D}(g) \wedge$$

$$(\exists f)(\exists \alpha)[\text{Ordinal}(\alpha) \wedge \text{s-alpha func}(f) \wedge$$

$$(\exists \beta < \alpha)(\langle p, f, g\rangle \in f_\beta)]$$

Lastly, we want a formula for $p \Vdash [\![f \approx g]\!]$ (currenlty we just have one for $p \Vdash f \approx g$). This is just

$$(\forall q \overleftarrow{\mathcal{R}}\, p)(\exists q' \overleftarrow{\mathcal{R}}\, q)(\text{Equal}(q', f, g))$$

and this formula satisfies the conditions of Proposition 5.3.

Theorem 5.4

There are multiple things to note about this theorem.

First, the proof given for the atomic case is incorrect. The formula should instead be

$$z \in \mathcal{G} \wedge \mathcal{D}(f) \wedge \mathcal{D}(g) \wedge$$
$$(\forall z' \overleftarrow{\mathcal{R}}\, z)(\exists z'' \overleftarrow{\mathcal{R}}\, z')(\text{Equal}(z'', w, x) \wedge$$
$$(\forall z''' \overleftarrow{\mathcal{R}}\, z'')(\exists z'''' \overleftarrow{\mathcal{R}}\, z''')(\langle z'''', w\rangle \in y))$$

Also, because we have the Equal(p, f, g) formula, it should be noted that Theorem 5.4 can be easily altered so that, if we extend the language $\mathcal{L}_c$ to include that $\approx$ relation (i.e., so that formulas can be built out of the $\approx$ relation) and the ϵ relation, the same result will follow.

The negation case is slightly incorrect. Instead, we should take the formula $F_\phi(z, x_1, ..., x_n)$ to be

$$(\forall z' \overleftarrow{\mathcal{R}}\, z)(\exists z'' \overleftarrow{\mathcal{R}}\, z') \sim F_\psi(z'', x_1, ..., x_n)$$

Lastly, here is the answer for Exercise 5.2. The last remaining case in this theorem is for the existential quantifier. Consider some formula $\phi(x_1, ..., x_n)$ of the form $(\exists x)\psi(x, x_1, ..., x_n)$. Recall that $[\![(\exists x)\psi(x, x_1, ..., x_n)]\!]$ is $\Box\Diamond(\exists x)[\![\psi(x, x_1, ..., x_n)]\!]$. Then take $F_\phi(z, x_1, ..., x_n)$ to be the formula

$$(\forall z' \overleftarrow{\mathcal{R}}\, z)(\exists z'' \overleftarrow{\mathcal{R}}\, z')(\exists x) F_\psi(z'', x, x_1, ..., x_n)$$

Notes

Here is an explanation for why g^* (as defined on page 242) is in $\mathcal{D}^\mathcal{G}$. We are assuming $g \in \mathcal{D}^\mathcal{G}$, so $g \in R^\mathcal{G}_\alpha$ for some α. All members of g^* are thus in $\mathcal{G} \times R^\mathcal{G}_\alpha$, meaning, if we can show that $g^* \in M$, then it will follow that $g^* \in R^\mathcal{G}_{\alpha^+}$, hence $g^* \in \mathcal{D}^\mathcal{G}$. Well, g^* is definable over M by

$$(\exists q)(\exists f)(x = \langle q, f\rangle \wedge (\exists p \mathcal{R} q)(\langle p, f\rangle \in g))$$

Thus g^* is a definable subset of $\mathcal{G} \times R^\mathcal{G}_\alpha$, meaning $g^* \in M$, and therefore, $g^* \in \mathcal{D}^\mathcal{G}$.

Proposition 6.1

Note why $F \in \mathcal{D}^\mathcal{G}$. Let $\psi(s)$ be the formulas

$$(\forall x)(x \in s \supset (\exists y)\phi(x, y))$$

Recall the notation used in Theorem 5.4. We have a formula $F_\psi(v, s)$, such that

$$p \vDash [\![\psi(f)]\!] \text{ iff } F_\psi(p, f) \text{ is true over} M$$

Now, because we are assuming $f \in \mathcal{D}^\mathcal{G}$, it easily follows that $F \subseteq \mathcal{G} \times R^\mathcal{G}_\alpha$ for some α. Thus, if we show that $F \in M$, it follows that $F \in R^\mathcal{G}_{\alpha^+}$, and thus also $\mathcal{D}^\mathcal{G}$. Well, F is definable over M by the formula

$$q \in \mathcal{G} \wedge (\forall q' \overleftarrow{\mathcal{R}}\, q)(\exists q'' \overleftarrow{\mathcal{R}}\, q')\langle q'', a\rangle \in f \wedge F_\psi(q, f)$$

meaning F is a first-order definable subset of $\mathcal{G} \times R^\mathcal{G}_\alpha$, and thus $F \in M$, hence $F \in \mathcal{D}^\mathcal{G}$.

Here is why Proposition 6.1 yields the validity of the translate of the collection schema over all $S4$ models of M. Suppose for some world p

$$p \nvDash [\![(\forall a)[(\forall x \in a)(\exists y)\phi(x, y) \supset (\exists b)(\forall x \in a)(\exists y \in b)\phi(x, y)]]\!]$$

Then, for some $f \in \mathcal{D}^{\mathcal{G}}$,

$$p \nVdash [\![(\forall x \in f)(\exists y)\phi(x,y) \supset (\exists b)(\forall x \in f)(\exists y \in b)\phi(x,y)]\!]$$

which means, for some p' with $p\mathcal{R}p'$, $p' \Vdash [\![(\forall x \in f)(\exists y)\phi(x,y)]\!]$ and (1) $p' \nVdash [\![(\exists b)(\forall x \in f)(\exists y \in b)\phi(x,y)]\!]$. By Proposition 6.1, it follows that there is some $g^* \in \mathcal{D}^{\mathcal{G}}$ such that $p' \Vdash [\![(\forall x \in f)(\exists y \in g^*)\phi(x,y)]\!]$, but this is in direct contradiction with (1).

Note that there is a familiar (yet inconsequential) error in this proof. It is claimed that $p' \Vdash [\![a \in f]\!]$ means for some a', $p' \Vdash [\![a \approx a']\!]$ and $p' \Vdash [\![a'\epsilon f]\!]$, but in reality, this means that for some p'' with $p'\mathcal{R}p''$ and for some a', $p'' \Vdash [\![a \approx a']\!]$ and $p'' \Vdash [\![a'\epsilon f]\!]$.

Exercise 6.1

Here is the full formula $\psi(u,v)$ used in the proof of Proposition 6.1 (using the notation introduced in section 5 of this chapter):

$$\mathcal{D}(u) \wedge \mathcal{D}(v) \wedge (\exists q \in \mathcal{G})(\exists q' \overleftarrow{\mathcal{R}} q)(\exists a)(\exists b)[$$
$$u = \langle q, a\rangle \wedge v = \langle q', b\rangle \wedge F_\phi(q', a, b)]$$

Exercise 6.2

Here is why $g \in \mathcal{D}^{\mathcal{G}}$. Because h is a set (in M), and g is a subset of it, we already know that $g \in M$ (the formula defining g over M is given in the book). Now, we know that $g \subseteq \mathcal{D}^{\mathcal{G}}$. So, if we can show that there is some $\mathcal{G}$-rank greater than all of the ranks of the elements of g, it would follow that $g \in \mathcal{D}^{\mathcal{G}}$. But suppose that wasn't the case; that is, that there is no $\mathcal{G}$-rank greater than all of the ranks of the elements of g. Then it is easy to see that the function assigning each element x of g to the $\mathcal{G}$-rank of x is first-order definable over M, and thus, because g is itself in M, the range of that function is in M. But, by assumption, there is no ordinal in M greater than all of the ordinals in the range of the function, and an easy consequence of this is that the union of the range (which is also in M) is an ordinal greater than all of the ordinals in M (or On itself, which is not a set), a contradiction. Thus it must be that there is some ordinal in M greater than all of the $\mathcal{G}$-ranks of the elements of g, meaning $g \in \mathcal{D}^{\mathcal{G}}$.

Section 7

Here is why the principle discussed in this section is equivalent to the Axiom of Choice (AC) (in the form that every non-empty set has a choice function). Call that principle (that every set f of non-empty and non-overlapping members has a choice set) Axiom S.

On the one hand, we show that Axiom S implies AC. Consider some non-empty set f. We must show that it has a choice function. For each subset S of f, let S' be $\{S\} \times S$. Then consider the set D of all S' (for each non-empty subset S of f). D is then a set of non-empty and non-overlapping sets (non-overlapping, as for any S_1' and S_2' in D, where $S_1 \neq S_2$, each member of S_1' is of the form $\langle S_1, x\rangle$ for $x \in S_1$, and each member of S_2' is of the form $\langle S_2, y\rangle$ for $y \in S_2$, and because $S_1 \neq S_2$, $\langle S_1, x\rangle \neq \langle S_2, y\rangle$ for all elements $x \in S_1$ and $y \in S_2$). Thus, by Axiom S, there is a choice set for S', and this is a choice function for f.

On the other hand, we show that AC implies Axiom S. Consider some non-empty set f (the case where f is empty is obvious) all of whose elements are non-empty and non-overlapping. We must show that there is a choice set for f. Consider $\cup f$. By AC, there is a choice function g for $\cup f$. Also, for each $x \in f$, $x \subseteq \cup f$. Then take s to be the set of all y where $g(x) = y$ for some $x \in f$. By the assumption that each element of f is nonoverlaping, s contains exactly one element of each $x \in f$.

There were numerous things that were unclear to me in this section. First, it is claimed that $s \in R^{\mathcal{G}}_{\alpha+1}$. Certainly, we know that $s \subseteq \mathcal{G} \times R^{\mathcal{G}}_\alpha$, yet I do not see why we know that $s \in M$, unless the well ordering of $R^{\mathcal{G}}_\alpha$ itself exists (so that we can say "$m(q,a)$ exists" in a formula defining s over M). Next, it was not clear to me why Propositions 7.8 and 7.10 yield

Theorem 7.11. It is clear to my why they yield the translate of S being valid over $\mathcal{M}$, where S is the formula that states that every set of non-empty, non-overlapping sets has a choice set. S is equivalent to AC *in the current model*, though this is not true in all classical models whatsoever (meaning I do not think we would be able to use Corollary 4.8 of Chapter 16 here).

Proposition 7.10

There is a familiar error (but again, inconsequential) in this proof: it is said that $p \vDash [\![a \in f \wedge x \in a \wedge x \in s \wedge y \in a \wedge y \in s]\!]$ implies that there are some a', x' and y' so that $p \vDash [\![a \approx a' \wedge x \approx x' \wedge y \approx y']\!]$ and $p \vDash [\![a' \epsilon f \wedge x' \epsilon a' \wedge x' \epsilon s \wedge y' \epsilon a \wedge y' \epsilon s]\!]$, but, in reality, it means that there is some p' with $p \mathcal{R} p'$ and some a', x' and y' so that $p' \vDash [\![a \approx a' \wedge x \approx x' \wedge y \approx y']\!]$ and $p' \vDash [\![a' \epsilon f \wedge x' \epsilon a' \wedge x' \epsilon s \wedge y' \epsilon a \wedge y' \epsilon s]\!]$.